

# Prime Factorization

Lesson 1-1

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

## Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*7 has only two factors:  $1 \times 7$ . So 7 is prime.*

### Prime number

A number bigger than 1 that you can only divide by 1 and itself.

*$12 = 1 \times 12, 2 \times 6, 3 \times 4$  — six factors, so 12 is composite*

### Composite number

A number bigger than 1 that you can divide by more than just 1 and itself.

*$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$*

### Prime factorization

Writing a number as prime numbers multiplied together.

*$24 \rightarrow 4 \times 6 \rightarrow (2 \times 2) \times (2 \times 3) \rightarrow 2 \times 2 \times 2 \times 3$*

### Factor tree

A picture that splits a number into its prime numbers, step by step.

*$2^3$  means  $2 \times 2 \times 2 = 8$*

### Exponent

A small number that tells how many times to multiply a number by itself.

## Key Ideas & Notes



### WHAT WE'RE LEARNING TODAY

**I can write a number as a product of its prime factors using a factor tree.**

#### 1 Notice

The space station received 60 supply crates.

#### 2 Key idea

I can write a number as a product of its prime factors using a factor tree.

#### 3 Words I'll use

Prime number

Composite number

Prime factorization

Factor tree

Exponent



#### Think About It

- What number are we breaking down into factors?
- What's the difference between a factor and a prime factor?
- How many different ways can we start breaking 60 apart?



#### Watch

Watch your teacher model one example. Jot what you see.

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#### We try

Solve the next one together as a class.

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#### You try

Now try one on your own.

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## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**Mission Control needs to break the 60 supply crates into prime parts. Is 60 a prime number or a composite number, and how can you tell before you even start the factor tree?**

#### Level 1 support

**Start here:** 60 is composite because it has more than two factors.



**Need a hint?**

**Sentence starters (optional)**

**60 is a \_\_\_\_\_ number because it has \_\_\_\_\_.**

*60 es un número \_\_\_\_\_ porque tiene \_\_\_\_\_.*

**I can tell before factoring because \_\_\_\_\_.**

*Puedo saberlo antes de factorizar porque \_\_\_\_\_.*

**Word bank:** **prime number** **composite number** **factor** **divisible** **factor tree**

**Listen for:** Students say 60 is composite because it has more than two factors (it is divisible by numbers other than 1 and 60, such as 2, 3, 5, 6), so it can be broken down.

#### Level 2

**Could Mission Control ever be asked to factor a prime number of crates, like 59?**

**Explain what its factor tree would look like and why.**

- A prime number like 59 \_\_\_\_\_ because its only factors are \_\_\_\_\_.
- Its factor tree would \_\_\_\_\_ because \_\_\_\_\_.

EXPLORE

Look at your factor tree for 60. How did you decide which numbers to break apart first, and how do you know when to stop?

Level 1 support

**Start here:** You keep splitting until every branch is a prime number.



Need a hint?

Sentence starters (optional)

I broke \_\_\_\_ into \_\_\_\_ because \_\_\_\_.

Separé \_\_\_\_ en \_\_\_\_ porque \_\_\_\_.

I knew I was finished when every factor was \_\_\_\_.

Supe que terminé cuando cada factor era \_\_\_\_.

A prime number is \_\_\_\_, so I cannot break it apart.

Un número primo es \_\_\_\_, así que no puedo separarlo.

Word bank: prime number composite number factor factor tree

prime factorization

**Listen for:** Students explain they keep splitting composite branches and stop only when every leaf is prime, ending with  $60 = 2 \times 2 \times 3 \times 5$ .

Level 2

Two classmates started 60 as  $6 \times 10$  and  $4 \times 15$ . Will they get the same prime factorization? Justify why or why not.

- They \_\_\_\_ get the same prime factorization because \_\_\_\_.
- No matter how I start the tree, I always end with \_\_\_\_ because \_\_\_\_.

CONNECT

**The party planner has 72 balloons. How could the prime factorization of 72 help her find every way to arrange the balloons in equal rows?**

Level 1 support

**Start here:** Combining the prime factors of 72 builds all of its factors.

 **Need a hint?**

**Sentence starters (optional)**

**The prime factorization of 72 is \_\_\_\_\_, written with exponents as \_\_\_\_\_.**

*La factorización prima de 72 es \_\_\_\_\_, escrita con exponentes como \_\_\_\_\_.*

**She can find all the factors by \_\_\_\_\_ the prime factors.**

*Puede hallar todos los factores al \_\_\_\_\_ los factores primos.*

**Word bank:** prime factorization factor exponent composite number product

**Listen for:** Students give  $72 = 2 \times 2 \times 2 \times 3 \times 3 = 2^3 \times 3^2$  and reason that multiplying different combinations of those primes produces all factors (1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72) and thus all equal-row arrangements.

Level 2

**Why does writing the prime factorization with exponents ( $2^3 \times 3^2$ ) make it easier to count how many factors 72 has than writing it the long way?**

- Exponents make it easier because \_\_\_\_\_.
- I can use the exponents to \_\_\_\_\_ instead of listing every factor by \_\_\_\_\_.

REFLECT

**How is a prime number different from a composite number, and why can every composite number be written as a product of primes?**

Level 1 support

**Start here:** Primes are the building blocks that every composite number is made from.

 **Need a hint?**

**Sentence starters (optional)**

**A prime number has \_\_\_\_, but a composite number has \_\_\_\_.**

*Un número primo tiene \_\_\_\_, pero un número compuesto tiene \_\_\_\_.*

**Every composite number can be built from primes because \_\_\_\_.**

*Todo número compuesto se puede formar con primos porque \_\_\_\_.*

**Word bank:** prime number composite number factor prime factorization  
product

**Listen for:** Students contrast exactly-two-factors (prime) with more-than-two-factors (composite) and explain that primes are the building blocks, so any composite splits into a unique product of primes.

Level 2

**Is 1 prime, composite, or neither? Defend your answer using the definition of a prime number.**

- 1 is \_\_\_\_ because it has \_\_\_\_ factor(s).
- It does not fit the definition of prime because \_\_\_\_.

## Guided Examples

### Example 1

**Which of the following is a prime number?**

**Solution:** 17 has exactly two factors: 1 and 17.  $15 = 3 \times 5$ ,  $21 = 3 \times 7$ , and  $9 = 3 \times 3$ , so they are all composite.

**Answer:** A. 17

### Example 2

**What is the prime factorization of 30?**

**Solution:**  $30 = 2 \times 15 = 2 \times 3 \times 5$ . All three factors (2, 3, 5) are prime, so  $2 \times 3 \times 5$  is the prime factorization.

**Answer:** A.  $2 \times 3 \times 5$

### Example 3

**What is the prime factorization of 18?**

**Solution:**  $18 = 2 \times 9 = 2 \times 3 \times 3$ . Both 2 and 3 are prime, so  $2 \times 3 \times 3$  is the prime factorization.

**Answer:** A.  $2 \times 3 \times 3$

## Write About the Math

### The Writing Revolution

I can explain how I broke a number down using the words prime number, composite number, factor, and exponent.

#### 1. Kernel Sentence subject + verb

**Model:** Prime factorization is writing a number as prime numbers multiplied together.  
*Factorización prima es escribir un número como números primos multiplicados.*

**Write a kernel sentence about prime factorization. Use a subject and a verb.**

*Escribe una oración base sobre factorización prima. Usa un sujeto y un verbo.*

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#### 2. Sentence Expansion because • but • so

**Kernel:** Prime factorization matters in math  
*Factorización prima importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Prime factorization matters in math because \_\_\_\_.**  
*Factorización prima importa en matemáticas porque \_\_\_\_.*

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**but**  
*pero*

**Prime factorization matters in math, but \_\_\_\_.**  
*Factorización prima importa en matemáticas, pero \_\_\_\_.*

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**so**  
*entonces*

**Prime factorization matters in math, so \_\_\_\_.**  
*Factorización prima importa en matemáticas, entonces \_\_\_\_.*

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### 3. Sentence Types 4 ways to write a math idea

#### Statement *Afirmación*

Tell one true fact about prime factorization.  
*Di un hecho verdadero sobre prime factorization.*

**Prime factorization** \_\_\_\_.

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#### Question *Pregunta*

Ask a question about prime factorization.  
*Haz una pregunta sobre prime factorization.*

**How does** \_\_\_\_ ?

*¿Cómo* \_\_\_\_ ?

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#### Exclamation *Exclamación*

Show excitement about prime factorization.  
*Muestra entusiasmo sobre prime factorization.*

**Wow,** \_\_\_\_ !

*¡Guau,* \_\_\_\_ !

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#### Command *Mandato*

Tell a partner what to do with prime factorization.  
*Dile a un compañero qué hacer con prime factorization.*

**First,** \_\_\_\_ .

*Primero,* \_\_\_\_ .

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### 4. Explain Your Reasoning use a sentence starter

**60 is a** \_\_\_\_ **number because it has** \_\_\_\_.

*60 es un número* \_\_\_\_ *porque tiene* \_\_\_\_.

**I can tell before factoring because** \_\_\_\_.

*Puedo saberlo antes de factorizar porque* \_\_\_\_.

**I broke** \_\_\_\_ **into** \_\_\_\_ **because** \_\_\_\_.

*Separé* \_\_\_\_ *en* \_\_\_\_ *porque* \_\_\_\_.

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## Try It

Solve on your own. Check the answer key when you are done.

**1. Which of these numbers is composite?**

- A. 27
- B. 23
- C. 29
- D. 31

Show your work:

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**2. Two students found different factor trees for 60. Student A started with  $2 \times 30$ . Student B started with  $6 \times 10$ . Which statement is true?**

- A. Both get the same prime factorization:  $2 \times 2 \times 3 \times 5$
- B. Only Student A gets the correct prime factorization
- C. Only Student B gets the correct prime factorization
- D. They will get different prime factorizations

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**Choose any two-digit composite number. Show TWO different factor trees that both lead to the same prime factorization. Explain why every composite number has only one prime factorization.**

*Sentence starter: I chose the number \_\_\_\_\_. My first factor tree starts with \_\_\_\_\_  $\times$  \_\_\_\_\_, and my second starts with \_\_\_\_\_  $\times$  \_\_\_\_\_. Both give the same prime factorization: \_\_\_\_\_. This happens because \_\_\_\_\_.*

Show your work:

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## Reflect — Exit Ticket

**What is the prime factorization of 40?**

- A.  $2 \times 2 \times 2 \times 5$
- B.  $4 \times 10$
- C.  $5 \times 8$
- D.  $2 \times 20$

Your answer:

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## Answer Key & Teacher Guide

1. **Try It 1:** A.  $27 - 27 = 3 \times 9 = 3 \times 3 \times 3$ , so it has more than two factors. 23, 29, and 31 are all prime.
2. **Try It 2:** A. Both get the same prime factorization:  $2 \times 2 \times 3 \times 5$  — *The Fundamental Theorem of Arithmetic says every composite number has exactly one prime factorization. No matter how you start the factor tree, you always end with  $2 \times 2 \times 3 \times 5$ .*
3. **Exit Ticket:** A.  $2 \times 2 \times 2 \times 5 = 40 = 2 \times 20 = 2 \times 2 \times 10 = 2 \times 2 \times 2 \times 5$ . All factors (2, 2, 2, 5) are prime.

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Prime factorization is writing a number as prime numbers multiplied together.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).